The remaining compound local groups in the Higman–Sims catalogue

Private L091/L096 finite component

We use the fixed 23-page supplement of Dimitri Leemans, dated 6 October 2003. Its diagram coordinates are those in the acquired source-coordinate ledger, not positions inferred from group orders. The reviewed local-data component supplies literal generators on 100 points for

$$B = \bigcap_i G_i, \qquad K_i = \bigcap_{j \neq i} \mathrm{Core}_{G_i}(G_i \cap G_j).$$

The earlier local-atom component identifies 350 of the 398 Borel and kernel occurrences. This note identifies the remaining 31 Borels and 17 kernels. It does not identify the full parabolics or their extensions over these kernels.

Theorem 1. *All 48 occurrences in* `source_joins.json` *have the displayed structures and the literal permutation models recorded there. Nine printed occurrences require correction, as specified on page 3. Together with the earlier 350 certificates this identifies all 86 Borels and all 312 residue kernels of the acquired catalogue.*

Finite transport and acquired names. The finite certificate gives images of the ordered generators of a named model M in an actual subgroup U. The verifier closes both permutation groups and constructs a bijection $f : M \to U$ fixing identity. For every element x and every model generator s it checks $f(xs) = f(x)f(s)$. Induction on positive words proves that f is a group isomorphism; inverses are positive powers because the groups are finite. This is the previously acquired generator-edge transport argument. The names of the reused subgroup-atlas models are acquired with their exact trees, degrees and generators. An order match alone is never used as a named isomorphism. New elementary factors are the literal cyclic, dihedral, quaternion, symmetric, alternating and affine models already proved in the first local-atom component. Direct products act on disjoint supports. The generalized dihedral model of order 50 acts on $\mathbb{F}_5^2$ by its two translations and $v \mapsto -v$.

Eight additional models have a recursive certificate $U = N : C_p$, with $p = 2$ or 3. A full literal map identifies N with an earlier model. The supplied element of order p lies outside N, normalizes it, and the verifier checks that the products nt^i are exactly U. Consequently the claimed split extension follows with its actual conjugation action. The recursive dependency order is checked; no model is justified by a circular reference. All maps and recursive factors are verified by `compound.py`, which makes no group-software or isomorphism call.

Reconciling different decompositions. A different extension tree does not by itself contradict a source label. We therefore supply ten additional normal-subgroup/complement certificates. Full generator-edge maps identify both factors, the normality relations are checked, their intersection is trivial, and their set product is the whole group. These certificates yield the following source-facing descriptions; $E_{p^d} = C_p^d$.

Base occurrence	Normal factor	Complement
2.2 Borel	C_4^3	S_4
3.6 Borel	E_8	S_4
3.10 Borel	E_{16}	A_4
3.20 Borel	E_{16}	C_2
3.24 Borel	Q_8	C_2
3.25 Borel	Q_8	C_2
4.2 Borel	E_{16}	E_4
4.7 Borel	E_{16}	C_2
4.12 Borel	E_4	E_4
4.11 Borel	E_9	C_2

For any second occurrence identified with the same exact model, compose its isomorphism with the inverse of the first. Transporting the factor generators gives the same decomposition in the second actual subgroup. The verifier checks these transported factors again, giving 32 factor checks at 28 distinct occurrences. In particular it pays the repeated $2^3 : S_4$, $2^4 : A_4$ and $2^4 : 2^2$ names. The order-40 Borel in 3.19 has an additional direct certificate $C_2 \times \mathrm{AGL}(1,5)$: the cyclic factors of orders 5 and 4 have the literal affine action, and the central involution is an independent factor.

The Borels in 3.25 and 3.26, and the boxed kernel in 3.26, are more precisely $Q_8 \times C_2$. They are split extensions written $Q_8 : 2$ in the source; the complement action is trivial. The Borel in 4.7 also splits. The source's dot is retained as extension notation here; we do not turn it into an unprinted nonsplitting assertion.

Lemma 1 (The common central kernel of order 64)**.** *The kernels in 2.2, type 2; 3.4, type 3; and 3.5, type 1 are central nonsplit extensions $C_4 \cdot E_{16}$.*

Proof. For the first actual kernel the certificate gives a central element z of order 4 and four quotient lifts. Their squares and commutators lie in $\langle z \rangle$. The sixteen binary words in these lifts give sixteen distinct cosets covering the group. The quotient is therefore E_{16}. Two supplied lifts do not commute. Multiplying either lift by a central element leaves their commutator unchanged; the verifier also checks all sixteen such pairs. An elementary abelian complement would provide commuting lifts of every two quotient elements, a contradiction. The other two kernels are explicitly isomorphic to this one by their common exact model. This proves the three printed $4 \cdot 2^4$ labels with the stronger central and nonsplit qualification. □

The nine incompatible printed occurrences. Let X denote the actual extraspecial group of order 32 in acquired model 9011, with its literal generators. Let R denote acquired model 21012, an extension $X \cdot E_4$ of order 128. The indicated actions, not just the extension shapes, are fixed by the serialized models.

Source occurrence	Printed label	Proved description
2.3 Borel	$4^3 : (7 : 3)$	$E_8 : L_2(7)$
3.4, 3.5 Borels	$(4^3 : 2^2) : 2$	$R : C_2$, order 256
3.4, type 2 kernel	$4^3 : 2^2$	$(((D_8 \times C_2) : C_2) : C_2) : C_2$, order 128
4.2, 4.3, 4.4, type 2 kernels	$2^4 : 2$	X
4.25 Borel and type 2 kernel	D_8	$C_4 \times C_2$

Here D_8 has order 8. These are nine separate entries in seven diagrams. The source coordinates and their literal model-type permutations are retained in the ledger.

For 2.3, four explicit commutators generate the actual group of order 1344, so it is nontrivial and perfect. The printed $4^3 : (7 : 3)$ is soluble: both its abelian normal factor and its group of order 21 are soluble. Thus the literal isomorphism to $E_8 : L_2(7)$ is a genuine correction, although the two displayed orders agree. For 3.4 and 3.5 the printed orders are respectively 256 for the kernel and 512 for the Borels; the actual certified orders are 128 and 256. The recursive positive models give structures as well as these order obstructions.

For each of the three order-32 kernels, transport identifies it with X. Direct checking gives $Z(X) = X' = C_2$, and every square lies in the centre. The commutator pairing on $X/Z(X) \cong \mathbb{F}_2^4$ is alternating and nondegenerate: a radical element would lift to the centre. If $A \leq X$ is abelian, $AZ(X)/Z(X)$ is isotropic, so has dimension at most 2. Hence $|A| \leq 8$. In particular X has no elementary abelian subgroup of order 16 and cannot have the printed $2^4 : 2$ structure. Finally, the two 4.25 groups have the explicit abelian model $C_4 \times C_2$, whereas D_8 is nonabelian.

Validation and scope. The new replay verifies all 48 occurrences and 48,647 generator edges, the stated factor certificates and the three obstruction arguments. Fourteen damaged certificates are rejected in both ordinary and optimized Python. Finite proposal calls and two unsuccessful factor attempts are preserved; proposals do not constitute the proof. One initially redundant commutator deletion was correctly accepted and therefore failed its intended negative control; the final control collapses all commutators instead. No ambient subgroup enumeration was repeated. The earlier two Borel corrections in 3.15 and 3.16 remain in the acquired 350-entry packet. Full parabolic and quotient structures, other source claims and whole-source registration are separate obligations.